Using AI Tools to Enhance the Sensory Afferent Database: A Comparative Study

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Abstract

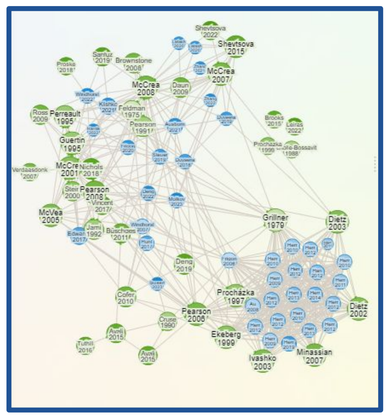
The Sensory Afferent Database represents a learning resource for researchers interested in understanding sensory afferent feedback in locomotion motor control. This was outlined in our previous work, “Sensory Afferent Database: Advancing towards a comprehensive understanding of sensory afferents and motor control”. The ongoing challenge lies in efficiently and accurately expanding this database with relevant literature, including both model-specific papers and comprehensive literature reviews. The integration of Artificial Intelligence (AI) tools promises a potential solution to this challenge, offering rapid processing capabilities.

Purpose

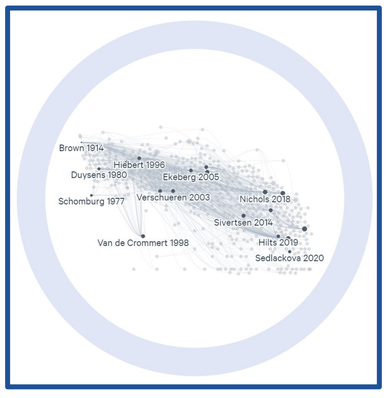
This study aims to evaluate the efficacy of current AI tools, such as Scispace and MirrorThink, in accurately and swiftly populating the Sensory Afferent Database. We investigate whether these tools can maintain a high standard of precision in identifying and summarizing pertinent research, thereby facilitating the database’s expansion. Our research questions focus on the reliability of AI-generated summaries and classifications compared to manual methods.

Methods

This study aims to evaluate the efficacy of current AI tools, such as Scispace and MirrorThink, in accurately and swiftly populating the Sensory Afferent Database. We investigate whether these tools can maintain a high standard of precision in identifying and summarizing pertinent research, thereby facilitating the database’s expansion. Our research questions focus on the reliability of AI-generated summaries and classifications compared to manual methods.



Research Rabbit showing current network connections in green and later work in blue.



Litmaps network connections.

Results

Our findings indicate that while AI tools exhibit the ability to process and summarize neuroscience papers, their reliability varies across different types of documents. For certain subsets of neuroscience literature, particularly those involving complex models or nuanced discussions, AI-generated outputs required significant human correction to ensure accuracy and relevance. However, for more straightforward literature reviews and model descriptions, AI tools performed adequately, reducing the time and effort needed for database population.

Table: Qualitative evaluation of each method as good (**+**) , adequate (**-**), or insufficient (**X**).

Tool	Identifying	Summarizing	Categorizing	Speed
Graduate Student	+	+	+	-
ResearchRabbit	+	+	X	+
Litmaps	-	X	X	X
txyz	X	+	-	+
SCISPACE	X	+	X	+
MirrorThink Empowering Thinkers	X	X	X	X

Conclusion

Integrating AI tools into the workflow of populating the Sensory Afferent Database offers a promising avenue for enhancing its growth. However, our study underscores the indispensable role of human oversight in ensuring the accuracy and relevance of AI-processed literature. The goal is to develop a hybrid model that leverages the speed and efficiency of AI while retaining the critical analytical depth that only human expertise can provide.

Next Steps

We will continue expanding the database by adding papers, models, review papers, and feedback types. The current entries will also be more fully filled out.

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